Hüseyin Felek

R&D Engineer | Industrial Automation & Process Instrumentation

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SUMMARY

Engineer with nearly 5 years in industrial automation and process instrumentation, carrying products and projects from requirement definition through design, testing, documentation, handover to production and first field trials. Owned a new product end to end, rolled out a redesigned controller architecture to 1000+ field devices, and coordinated with customers, production, quality and field service teams, and an international technology vendor. Hands-on with PLC and industrial communication (CODESYS, PROFINET, Modbus), production testing and quality assurance, and project tracking in Jira and Confluence.

WORK HISTORY

Software Design Engineer · Andasis Elektronik 05.2025 – 02.2026

- Delivered core network features (VLAN, IEEE 802.1X/RADIUS, ACL) for an OpenWrt-based switching and routing platform to production readiness, troubleshooting reliability issues at packet level with Wireshark.

Embedded Systems R&D Engineer · Eliar Elektronik 07.2021 – 05.2025

Project and Product Delivery

- Defined requirements and specifications for a new dual-channel pH/ORP product based on customer needs and owned it end to end, coordinating design, firmware, documentation and certification activities through to production readiness.
- Redesigned the I/O architecture of the company's automation controllers and rolled it out to 1000+ field devices, replacing legacy controllers.
- Coordinated integration with an international PROFINET stack vendor, managing technical alignment and issue resolution.
- Took part in the import process for pH probes sourced from abroad, supporting the delivery of components needed for production.
- Defined compliance tests and ran certification testing for the company's first officially BLE-certified device.
- Tracked tasks and maintained project documentation in Jira and Confluence.

Handover, Testing and Quality

- Defined production test procedures, manufacturing guidelines and post-production quality criteria, and handed production tests and calibration procedures over to the production team.
- Built CODESYS-based and PC-based test and calibration stations with automated test sequencing and data logging, shortening test and calibration time in series production.
- Took part in first field trials of new devices, carried out post-delivery checks and resolved defects found after delivery; led post-production quality assurance for the controller platform.

Customer and Field Interface

- Led a 6-month on-site feasibility study at customer dyehouses: interviewed process teams, collected samples, analyzed data with Python, SQL and Power BI, and presented the findings to management.
- Served as the technical escalation point for customer and field issues, running root cause analysis and responding to customers directly; trained the field service team on calibration procedures.

Automation and Industrial Communication

- Architected an embedded Linux automation controller platform with software-defined variants, from a PROFINET I/O coupler to a full CODESYS PLC.
- Developed IEC 61131-3 applications in CODESYS and device drivers for I/O module data exchange over Modbus and PROFINET.

CORE COMPETENCIES

Project Coordination	Requirement and scope definition, task and schedule tracking (Jira), vendor coordination, customer interface, project documentation (Confluence), Scrum
Testing & Quality	Production test procedures, test and calibration stations, quality criteria, certification testing, root cause analysis
Industrial Automation	PLC programming (CODESYS, IEC 61131-3), PROFINET, Modbus RTU/TCP, IO-Link, 4-20 mA, process instrumentation (pH/ORP, conductivity, optical)
Office & Data Tools	MS Office (Excel, PowerPoint), Power BI, SQL, Python
Software	C/C++, embedded Linux, Docker, Git

EDUCATION & LANGUAGES

B.Sc., Electrical and Electronics Engineering · Boğaziçi University 2016 – 2021

Turkish Native | English Advanced | French Intermediate

Driving licence Class B | Willing to travel internationally